Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. Students should take the signature from the faculty in charge. For every laboratory session, the observation component carries marks

1. Traffic Signal Driver Action

Problem

Design a flowchart that checks the current color of a traffic light and tells a driver what to do:

- If the light is Red, display "Stop".
- If the light is Yellow, display "Slow Down".
- If the light is Green, display "Go".

Solution

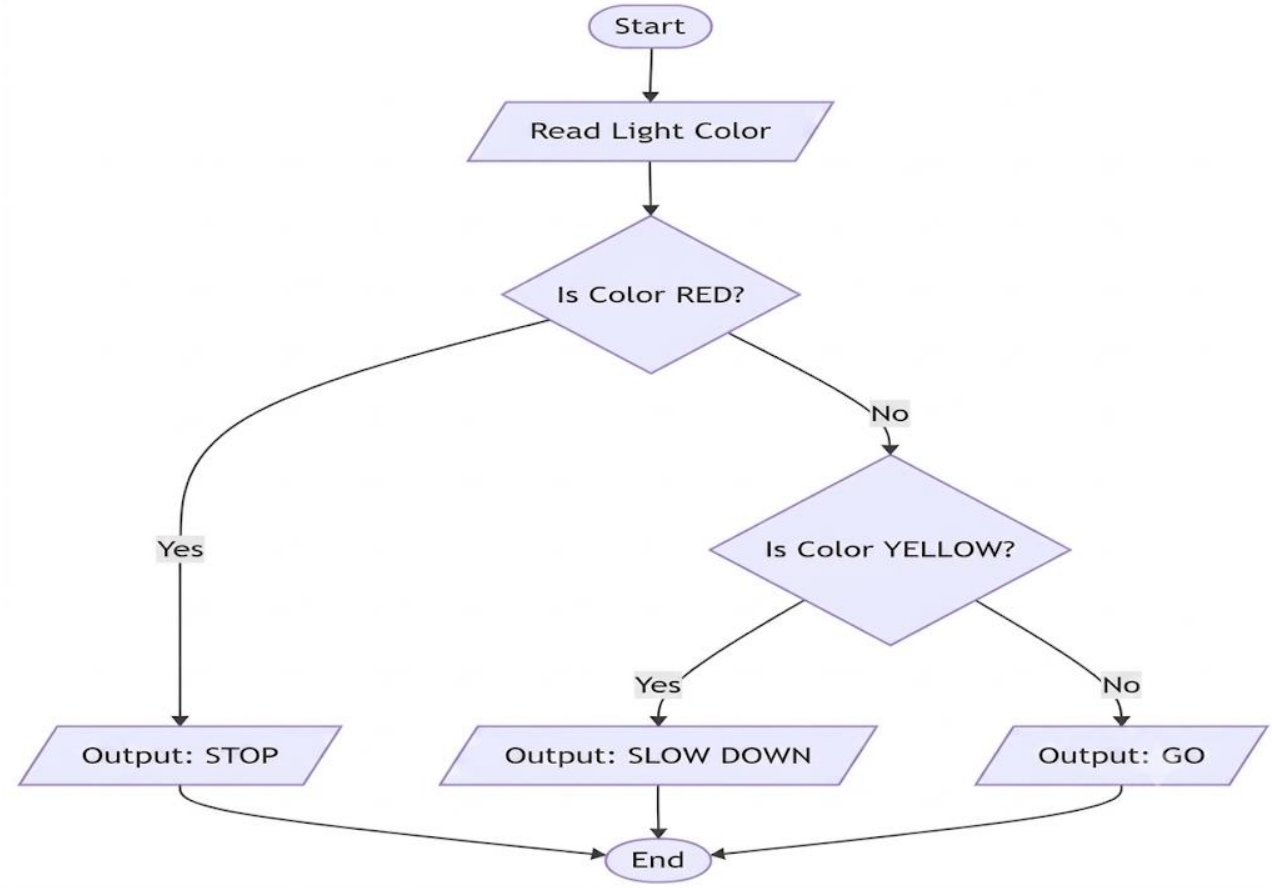
1. Read the `Light Color` input.

2. Check if `Color == Red` . If Yes, output "Stop".

3. If No, check if `Color ==Yellow` . If Yes, output "Slow Down".

4. If No, output "Go".

Flowchart



2. Exam Pass / Fail Checker

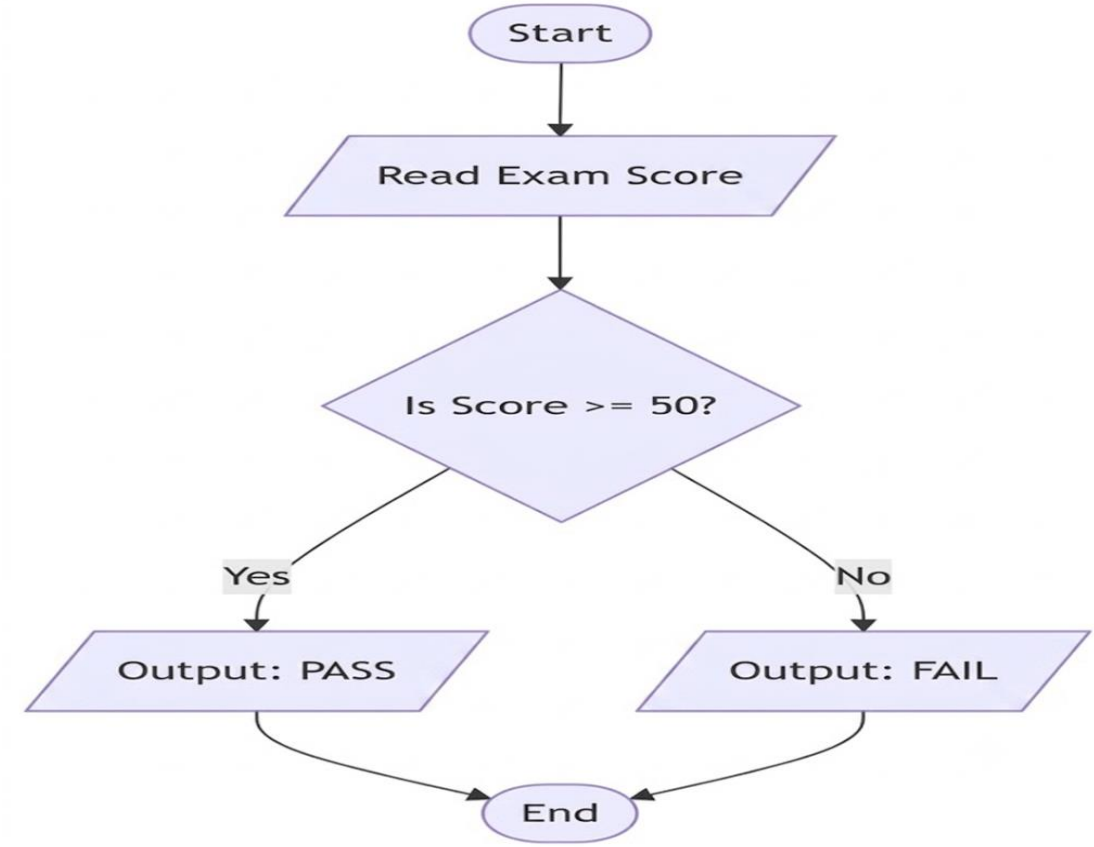
Problem

A school considers a student to have passed an exam if their score is 50 or higher out of 100. Draw a flowchart that reads a student's score and determines if they "Passed" or "Failed".

Solution

1. Read the student's `Score`.
2. Compare `Score>=50`.
3. If True, display "Pass".
4. If False, display "Fail".

Flowchart



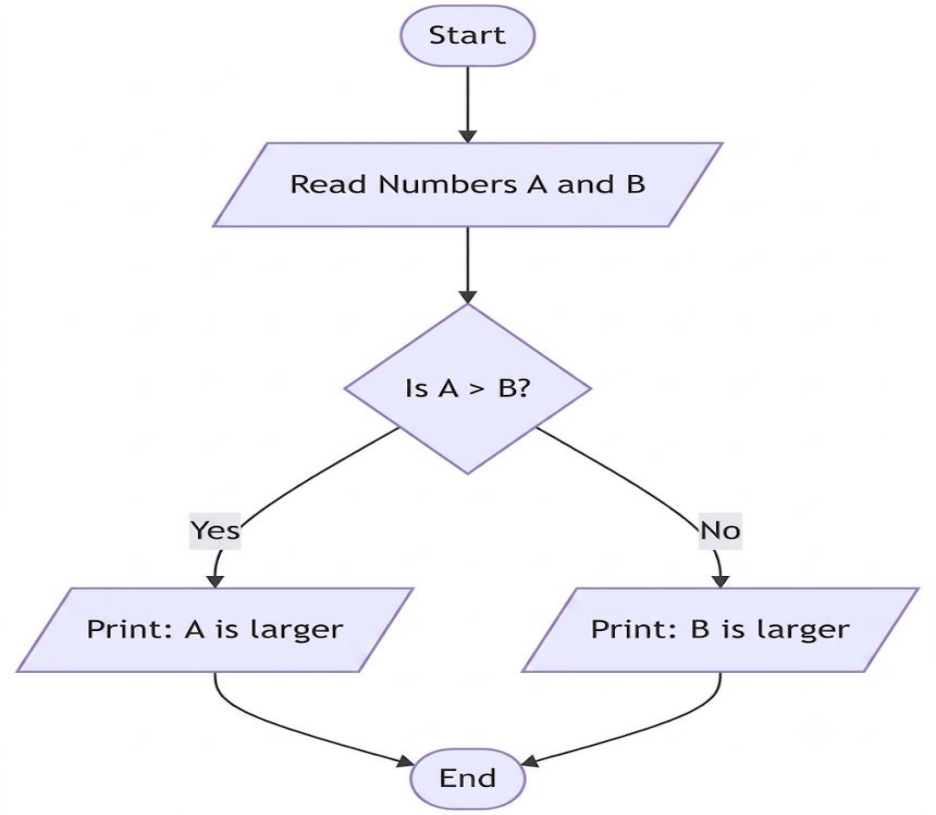
3. *Finding the Larger of Two Numbers*

Problem

Create a flowchart that takes two numbers, A and B, compares them, and prints which number is larger. (Assume the two numbers are not equal). Solution

1. Read inputs A and B.
2. Check if $A > B$.
3. If Yes, print "A is larger".
4. If No, print "B is larger".

Flowchart



4. Simple Purchase Approval

Problem

An online store needs to check if a customer has enough money to buy an item.

- Input the item's `Price` and the customer's `Account Balance`.
- If`Balance >= Price` , approve the purchase and subtract the price from the balance.
- If `Balance < Price` , display "Insufficient Funds".

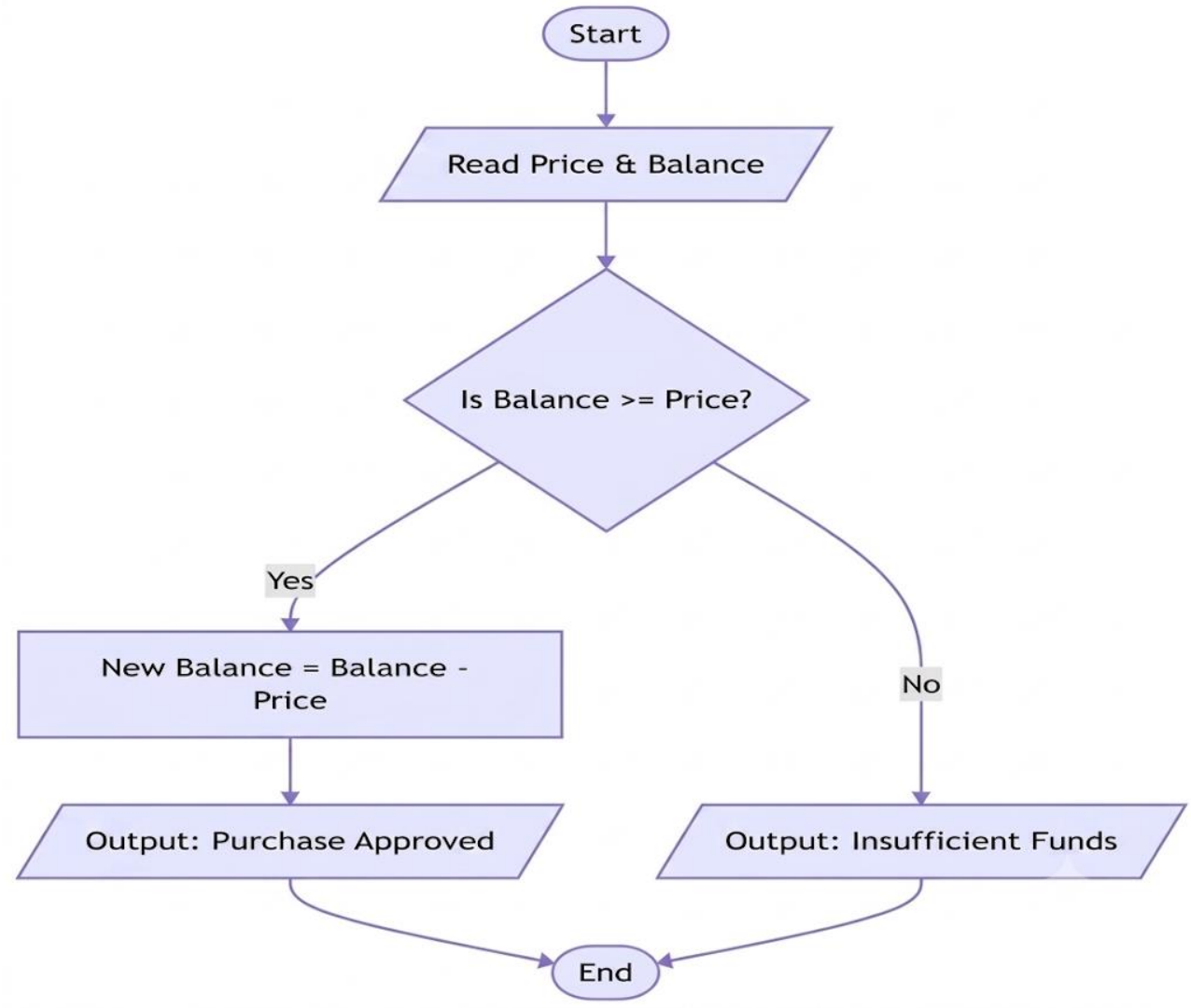
Solution

1. Read `Price` and `Balance`.

2. Check if `Balance>=Price`.

3. If Yes, calculate NewBalance = Balance - Price , display "Purchase Approved", and show the new balance.
4. If No, display "Insufficient Funds".

Flowchart



5. Self-Checkout Discount & Tax Calculator Problem

A retail self-checkout counter calculates final bill totals based on customer loyalty status and purchase total (□):

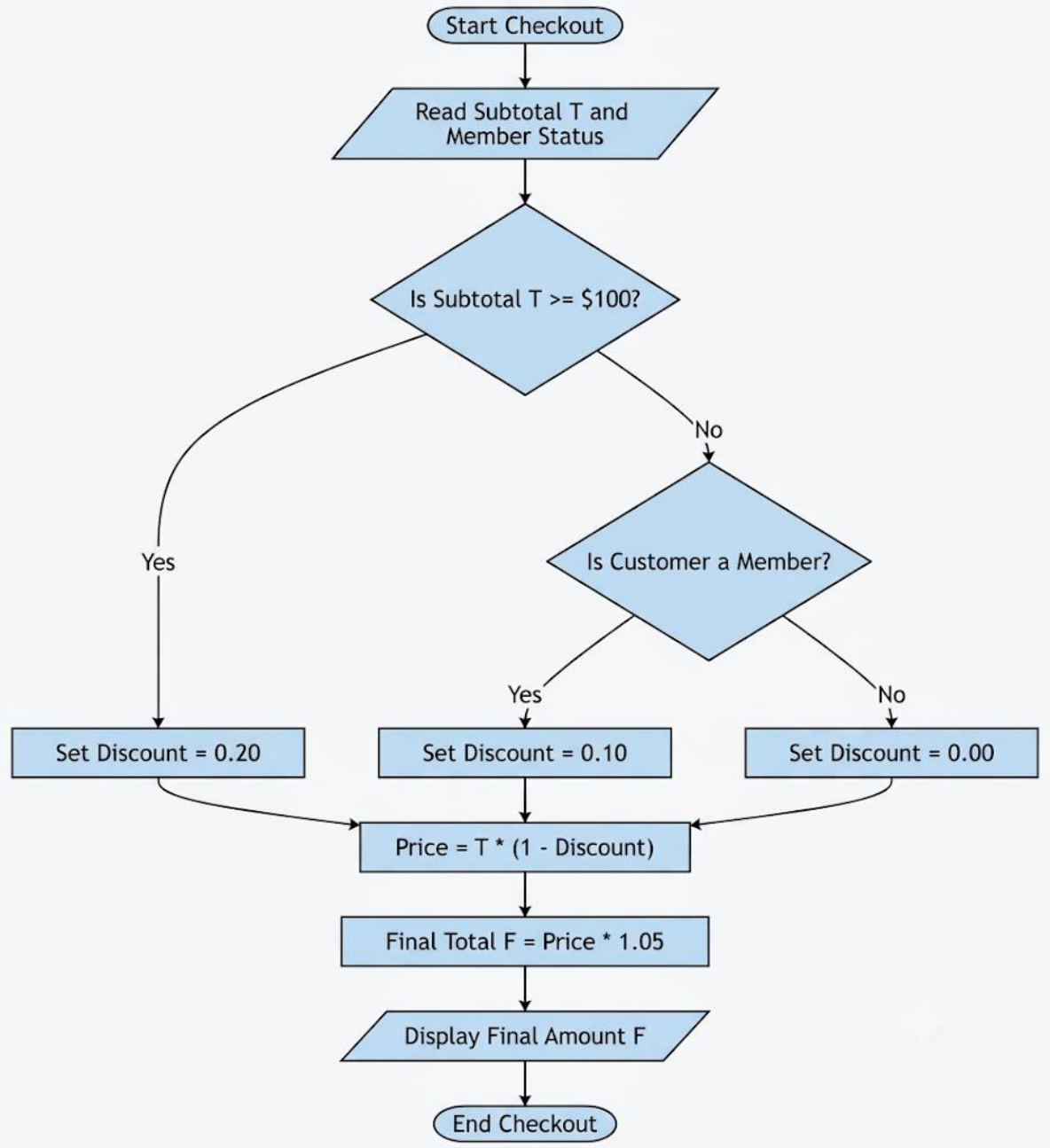
- If □≥$100, apply a ***20% discount***.
- If □<$100 AND customer is a ***Store Member***, apply a ***10% discount***.
- Otherwise, apply ***0% discount***.
- After applying discount, add a ***5% sales tax*** to calculate the final total □.

Question: Construct a flowchart to calculate final payment amount □.

Solution

1. Read subtotal □ and IsMember status (True/False).
2. If □≥100, set Discount = 0.20.
3. If □<100, check IsMember. If True, Discount = 0.10; if False, Discount = 0.00.
4. Calculate discounted price: □□□□□=□×(1−□□□□□□□□□).
5. Calculate final price with tax: □=□□□□□×1.05.
6. Display final amount □.

Flowchart


Start Checkout
Read Subtotal T and Member Status
Is Subtotal T >= $100?
No
Yes
Is Customer a Member?
Yes
No
Set Discount = 0.20
Set Discount = 0.10
Set Discount = 0.00
Price = T * (1 - Discount)
Final Total F = Price * 1.05
Display Final Amount F
End Checkout

Laboratory Week 1

6. Hospital Patient Triage Classification Problem

An automated triage workstation reads a patient's ***Oxygen Saturation (□₂ in %)*** and ***Heart Rate (□□ in bpm)*** to assign medical urgency:

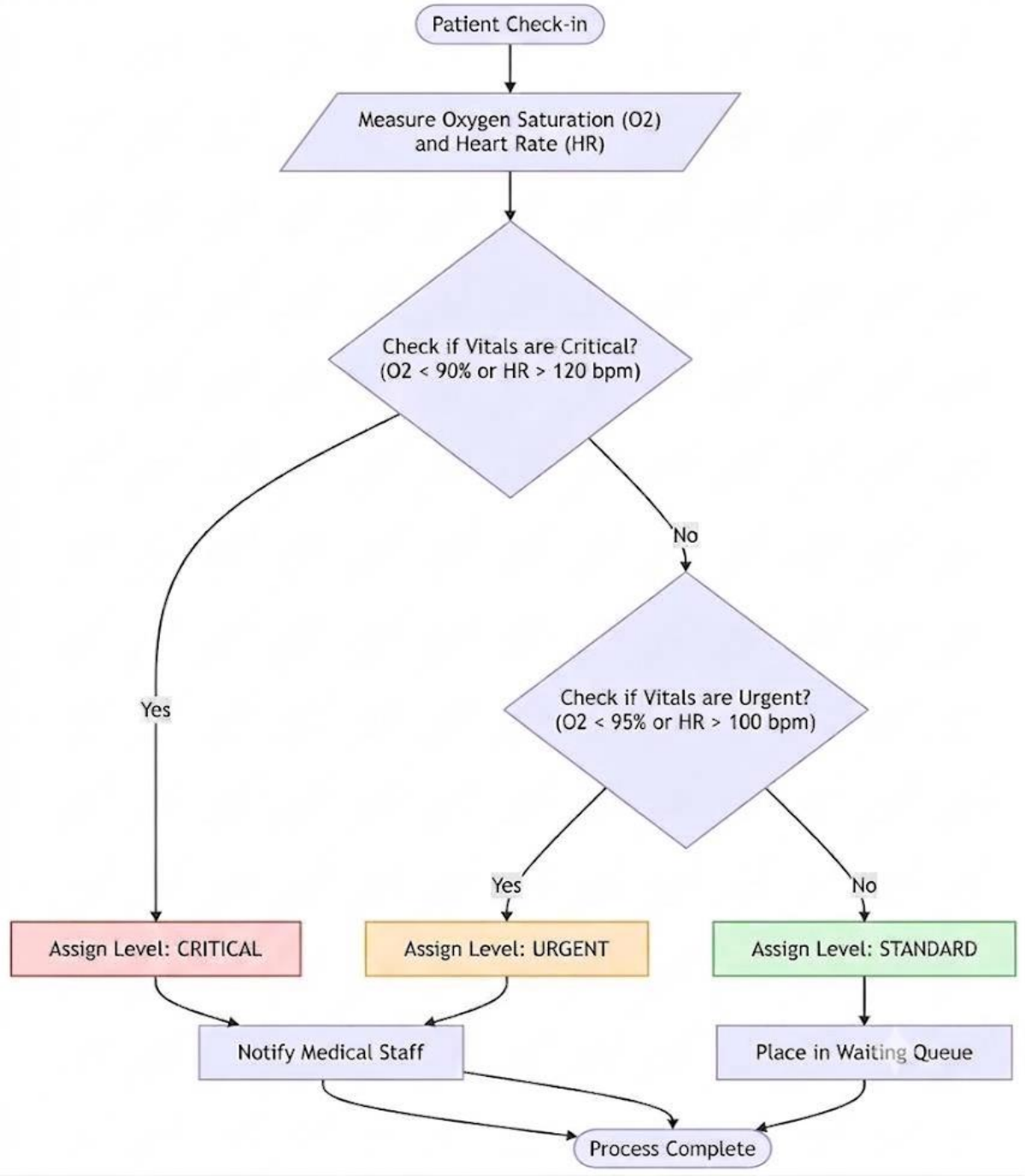
- ***Critical:*** If □₂<90% OR □□>120 □□□.
- ***Urgent:*** If NOT Critical, but □₂<95% OR □□>100 □□□.
- ***Standard:*** If □₂≥95% AND □□≤100 □□□.

Question: Draw a flowchart to classify patients into Critical, Urgent, or Standard priority.

Solution

1. Read □₂ saturation and □□ values.
2. Evaluate condition 1: Is □₂<90 OR □□>120? o If ***Yes***, assign ***Critical Priority***.
3. If ***No***, evaluate condition 2: Is □₂<95 OR □□>100?
    - o If ***Yes***, assign ***Urgent Priority***.
    - o If ***No***, assign ***Standard Priority***.

Flowchart


Patient Check-in
Measure Oxygen Saturation (O2) and Heart Rate (HR)
Check if Vitals are Critical? (O2 < 90% or HR > 120 bpm)
Yes
No
Check if Vitals are Urgent? (O2 < 95% or HR > 100 bpm)
Yes
No
Assign Level: CRITICAL
Assign Level: URGENT
Assign Level: STANDARD
Notify Medical Staff
Place in Waiting Queue
Process Complete

7. Smart Thermostat Eco Mode Logic Problem

A smart home thermostat controls heating/cooling based on a motion sensor and target temperature:

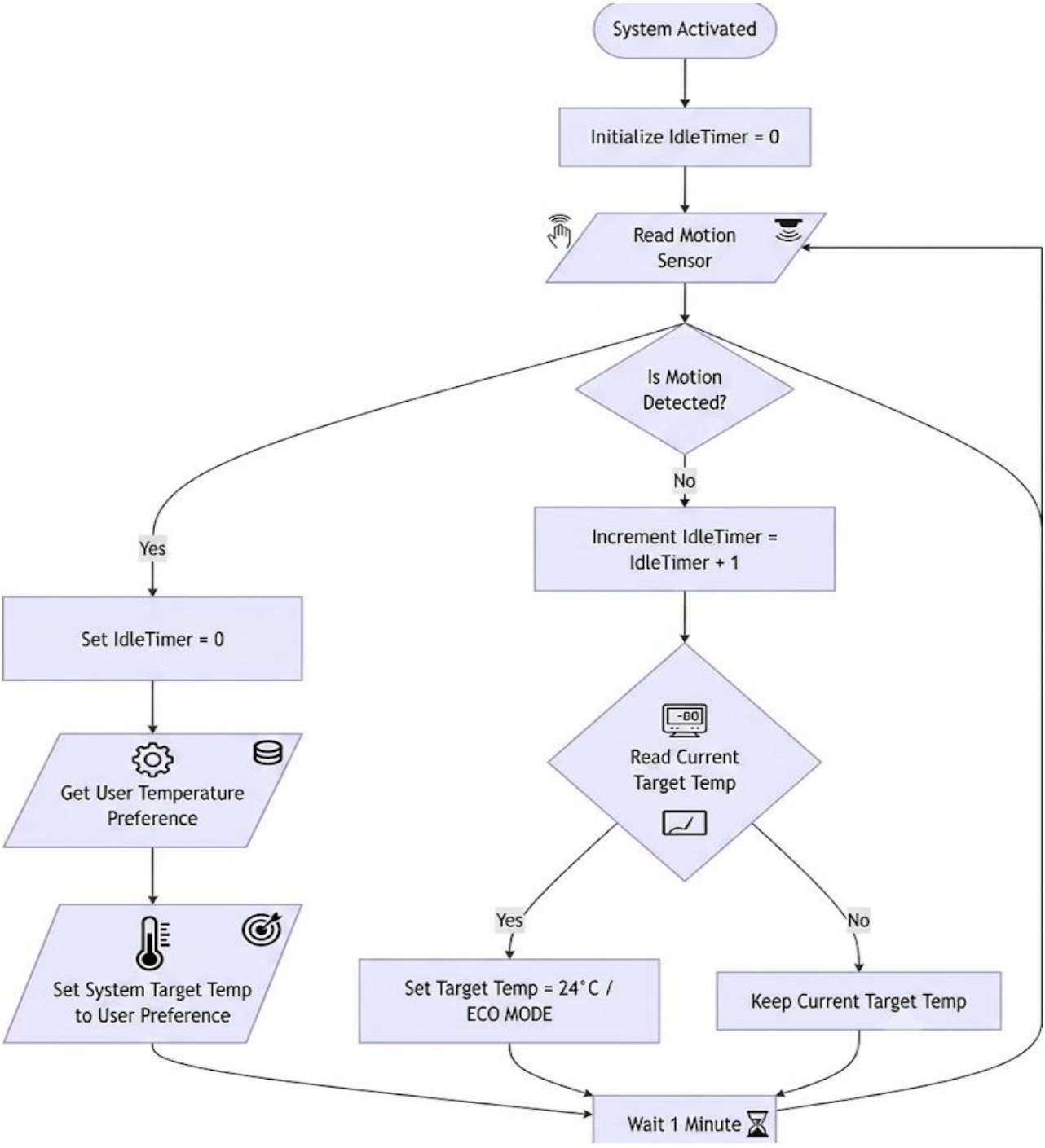
- Check motion sensor every minute.
- If ***Motion Detected == True***, set target temperature to user preference T_{pref} and reset IdleTimer = 0.
- If ***Motion Detected == False***, increment IdleTimer = IdleTimer + 1.
- If IdleTimer >= 30 mins, activate ***Eco Mode*** (Set target temperature to 24°C).

Question: Create a flowchart showing the loop logic for maintaining Eco Mode vs User Comfort Mode.

Solution

1. Initialize IdleTimer = 0.
2. Check motion sensor.
3. If motion is detected, set IdleTimer = 0, set $T_{target}=T_{pref}$. 4. If motion is not detected, increment IdleTimer = IdleTimer + 1.
5. Check if IdleTimer >= 30. ○ If ***Yes*** , set $T_{target}=24$ (Eco Mode).◦ o If ***No***, maintain current T_{target}.
6. Wait 1 minute and repeat check.

Flowchart


System Activated
Initialize IdleTimer = 0
Read Motion Sensor
Is Motion Detected?
No
Yes
Increment IdleTimer = IdleTimer + 1
Set IdleTimer = 0
Read Current Target Temp
Get User Temperature Preference
Yes
No
Set System Target Temp to User Preference
Set Target Temp = 24°C / ECO MODE
Keep Current Target Temp
Wait 1 Minute

8. Parking Garage Entry Barrier Gate Problem

An automated parking entry system manages available spots (□) out of a maximum capacity:

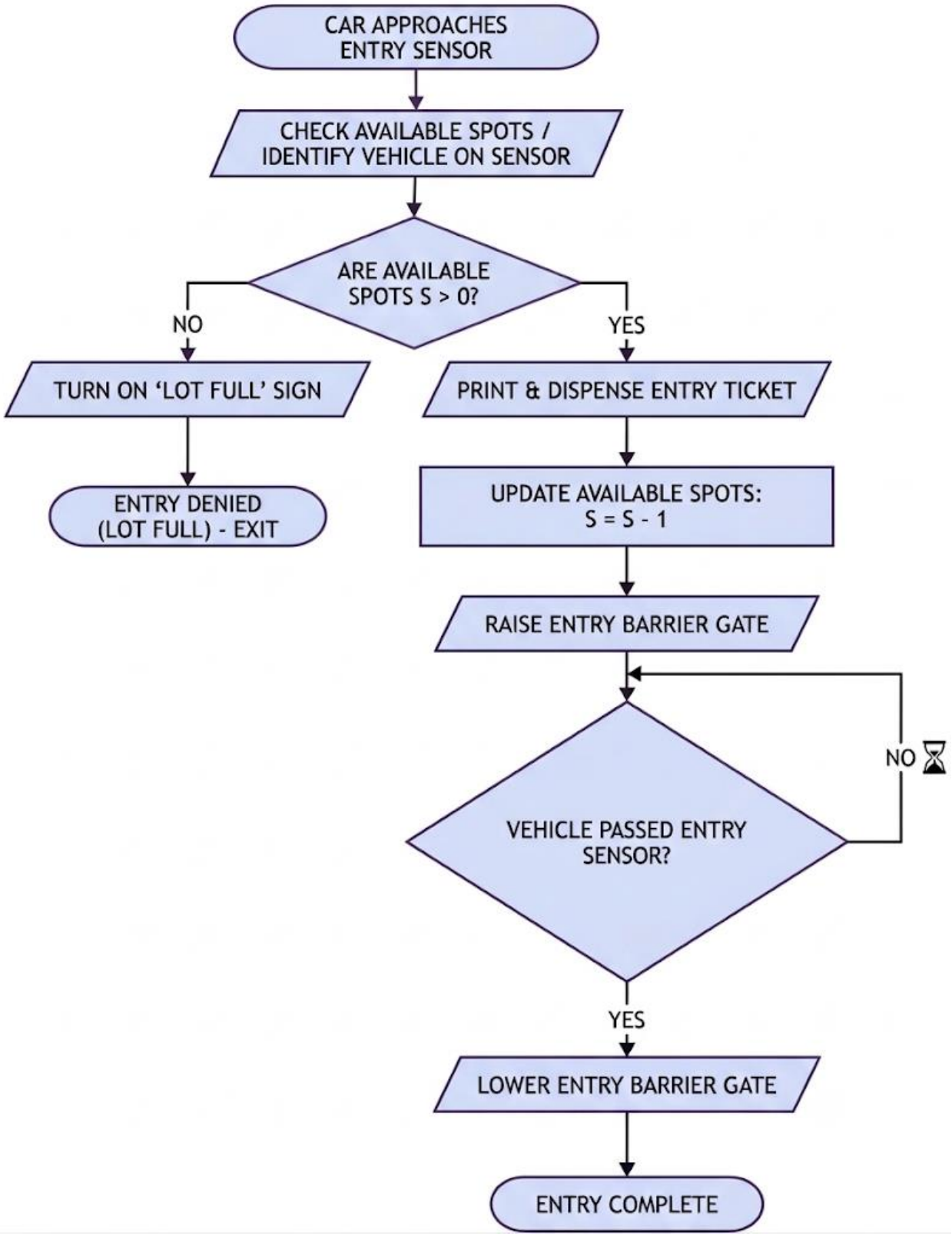
- When a vehicle approaches the entry sensor:
  - Check if available spots □>0.
  - If □>0, issue a parking ticket, decrement □=□−1, raise barrier gate, wait for car to cross exit sensor, then lower barrier gate.
  - If □==0, turn ON "LOT FULL" display sign and keep barrier gate closed.

Question: Design a flowchart to model the entry gate controller.

Solution

1. Detect vehicle at entrance sensor.
2. Check if □>0.
3. If □==0, display "LOT FULL" message and keep gate closed.
4. If □>0, print entrance ticket.
5. Decrement spot counter: □=□−1.
6. Raise entry gate barrier.
7. Monitor passing sensor until vehicle enters.
8. Lower entry gate barrier.

Flowchart


CAR APPROACHES ENTRY SENSOR
CHECK AVAILABLE SPOTS / IDENTIFY VEHICLE ON SENSOR
ARE AVAILABLE SPOTS S > 0?
NO
YES
TURN ON 'LOT FULL' SIGN
ENTRY DENIED (LOT FULL) - EXIT
PRINT & DISPENSE ENTRY TICKET
UPDATE AVAILABLE SPOTS: S = S - 1
RAISE ENTRY BARRIER GATE
VEHICLE PASSED ENTRY SENSOR?
NO
YES
LOWER ENTRY BARRIER GATE
ENTRY COMPLETE

Week 1

Prepared By, Devna
Sumesh